

MODULE ASSESSMENT 2

Week 2-3 Practical Task-Based Examination - Answer Sheet

Student ID: 2026CS012

Student Name: Aashna Shaikh

Department: CO

Date: 30 June 2026

Start Time:

End Time:

Answer Sheet Pages: 3

Invigilator Signature:

SUBMISSION LINK RECORD

Evidence type	Link submitted in portal	Access checked
Scanned answer-sheet PDF		
GitHub repository		
Colab notebook		
Jupyter notebook / Drive folder		
Other evidence		

STUDENT DECLARATION

I confirm that the handwritten work and practical notebook submitted for this assessment are my own work. I understand that inaccessible links, copied work, or missing evidence may affect evaluation.

Student Signature

Student Coordinator Check

Program Coordinator Review

SECTION A: SYSTEM DESIGN AND ALGORITHM

Workspace Flow:

VS code → Google Colab → Github Repository
 → Google Drive → student Portal Submission.

Explanation:

- 1> Vs code - Used to write, edit code with syntax highlighting and debugging.
- 2> Google colab - Useful if local Python is not installed. Runs Jupyter Notebook online.
- 3> Github Repository - Stores project files online. Tracks versions of code.
- 4> Google Drive - Stores handwritten PDF and notebook files used to share links.

5> student Portal - final place to submit all links and files.

CONTINUATION SHEET

Use additional copies if required

SECTION B: PYTHON FUNDAMENTALS

Q.3

 $a = 25$ a stores the value 25 $b = a$ b refers to the same value as a .So, $b = 25$ $c = 25$

A new variable is created for value 25

So, All three variables a, b, c points to the same value. $a = b = c = 25$

PRACTICAL WORK NOTES / REPOSITORY PLAN

STUDENT COORDINATOR VERIFICATION

PDF readable and page order correct	Yes / No
GitHub repository accessible	Yes / No
Notebook accessible and executed	Yes / No
Forwarded to Program Coordinator	Yes / No
Note	

Student Signature

SC Signature

PC Signature

on A.

Algorithm: Student Performance Analytics Pipeline

Input:

- student dataset containing student ID, Name, Attendance Percentage, Diary Entries, Tasks completed, Assignment score, and Lab score.

Process:

1. Start

2. Read the student dataset

3. Check whether the dataset is empty

- If the dataset is empty, display "No data available" and stop.

4. Check each record for missing values

- If any value is missing, report the missing data or skip that record.

5. For each student in the dataset:

- Read attendance, diary entries, tasks completed, assignment score, and lab score.

- Calculate the Readiness score using a suitable formula:

6. Classify the student based on the Readiness Score:

- Score is less than 40, Score is less than 70, Score is less than 90.

7. Repeat the above steps for all students.

8. Calculate the average performance of each department.

9. Find the student with the highest Readiness Score.

10. Prepare the list of students who need academic support.

11. Display the following insights:

- Department-wise average performance
- Top-performing student
- Students needing academic support.

12. Stop.

- Output:
- Readiness score of every
 - Performance Band (Support, Developing, Strong, or Excellent).
 - Department average performance
 - Top Performer.
 - List of students needing academic support.

Edge cases:

- Empty datasets
- Invalid attendance or marks (less than 0 or greater than 100).

Q.3

→ `names = ["Asha", "Riya"]`

In 'names' variable a list is stored.

`same_names = names`

The 'same_names' points to the same object as 'names'.

`same_names.append("Meha")`

`append()` is used to add a string at the end of list and it changes both the variables.

Final List:

`["Asha", "Riya", "Meha"]`

Q.4.

→ We'll check the condition one by one:

Iteration 1: Score = 38

score = 38

Is $38 < 40$? → Yes

band = support

Iteration 2: score = 62

Is $62 < 40$? → No

Is $62 < 70$? → Yes

band = Developing

Iteration ³ ~~Score~~: Score = 81

- Is $81 < 40$? $\rightarrow$ No
- Is $81 < 70$? $\rightarrow$ No
- Is $81 < 90$? $\rightarrow$ Yes

band = strong

Iteration 4: Score = 94

- Is $94 < 40$? $\rightarrow$ No
- Is $94 < 70$? $\rightarrow$ No
- Is $94 < 90$? $\rightarrow$ No

Else executes

band = Excellent

Q.5

$\rightarrow$

```
def readiness_score(attendance, diary_entries, task_count):  
    score = (attendance * 0.6) + (diary_entries * 2) + (task_count * 3)
```

```
    if score > 100:
```

```
        score = 100
```

```
    elif score < 0:
```

```
        score = 0
```

```
    else:
```

```
        score = score
```

```
    return score
```